

When 3% means nothing: calibrating escalation limits to a bank's own forecasting error distribution.

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Motivation

NII FORECASTING AND IRRBB GOVERNANCE

- Forecast accuracy for net interest income (NII) and interest-rate risk in the banking book (IRRBB) is central to earnings stability, balance-sheet management and supervisory credibility.
- Many institutions govern forecast performance with fixed deviation thresholds (3/4/5%), even though forecast uncertainty widens with the horizon and the error process may be heavy-tailed. A flat limit therefore carries no consistent probabilistic interpretation.
- The paper develops an integrated, probability-coherent framework for monitoring NII forecast errors and assessing IRRBB limit breaches.

Horizon-specific, quantile-anchored thresholds

THE SEP FAN-CHART LOGIC APPLIED TO ESCALATION LIMITS

Following the Federal Reserve's RMSE-based communication of forecast uncertainty, thresholds are anchored to the horizon-specific error scale:

$$C_{h,q} = z_q \cdot RMSE(h), \quad z_{80} = 0.84, \quad z_{90} = 1.28, \quad z_{95} = 1.64, \quad z_{99} = 2.33$$

under the Normal/RMSE mapping used for communication, with empirical quantiles replacing $z_q RMSE(h)$ where the error history warrants it.

- A standardized exceedance score expresses each realized error as a proportion of the 95% boundary, giving dashboards one common unit across horizons.
- Validation: unconditional and conditional coverage tests on the resulting interval forecasts, quantile elicibility, and a bias-dispersion decomposition separating structural misspecification from random variation.

The outer tier: extreme value theory

POT-GPD CALIBRATION OF RARE MISSES

- Because NII forecast errors may be heavy-tailed, the outer escalation tier is calibrated by peaks-over-threshold: by the Pickands–Balkema–de Haan theorem, excesses over a high threshold converge to a Generalized Pareto Distribution.
- POT-GPD modelling stabilises the outer tier against rare but severe misses, in analogy to Expected-Shortfall extensions of the VaR traffic-light backtest.
- The IRRBB extension quantifies the probability that limits on ΔNII are breached solely due to forecast or model uncertainty, mapping model risk into risk-appetite language.

Empirical illustration and implications

SIMULATION, N=120 PER HORIZON AT 3M/6M/12M/24M

- A transparent simulation reports empirical 90%/95% thresholds and EVT-based 97.5% outer-tier limits per horizon, demonstrates their stability, and contrasts them with flat limits, which over-penalise long-horizon errors and under-weight short-horizon ones.
- Comparability is restored across horizons because dispersion growth is embedded in the limit definition; conditional-coverage testing detects clustering of exceptions; breach probabilities for ΔNII are suitable for ALCO reporting.
- Limits of the approach: calibration is data-hungry at long horizons, and regime or model changes can break stationarity, requiring re-baselining, shadow backtests and, at adoption, wider bands or pooled-horizon estimation.

Thank you.

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